

Part I

SUMMARY of the Definition of the Derivatives

Summary

February 21, 2026

Summary

- The *average rate of change*, $AvRCh$, of a function f across an interval $[a, b]$ is given by

$$\begin{aligned} AvRCh &= \frac{\Delta f}{\Delta x} \\ &= \frac{f(b) - f(a)}{b - a} \end{aligned}$$

- The slope, m_{sec} , of the line secant to the graph of a function f at the points $(a, f(a))$ and $(b, f(b))$ is given by

$$\begin{aligned} m_{sec} &= \frac{\Delta y}{\Delta x} \\ &= \frac{f(b) - f(a)}{b - a} \end{aligned}$$

- The *derivative* of the function f at a , denoted $f'(a)$, is given by

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

and represents the instantaneous rate of change of f with respect to x at $x = a$ and also the slope of the line tangent to the graph of $y = f(x)$ at the point $(a, f(a))$.

Part II

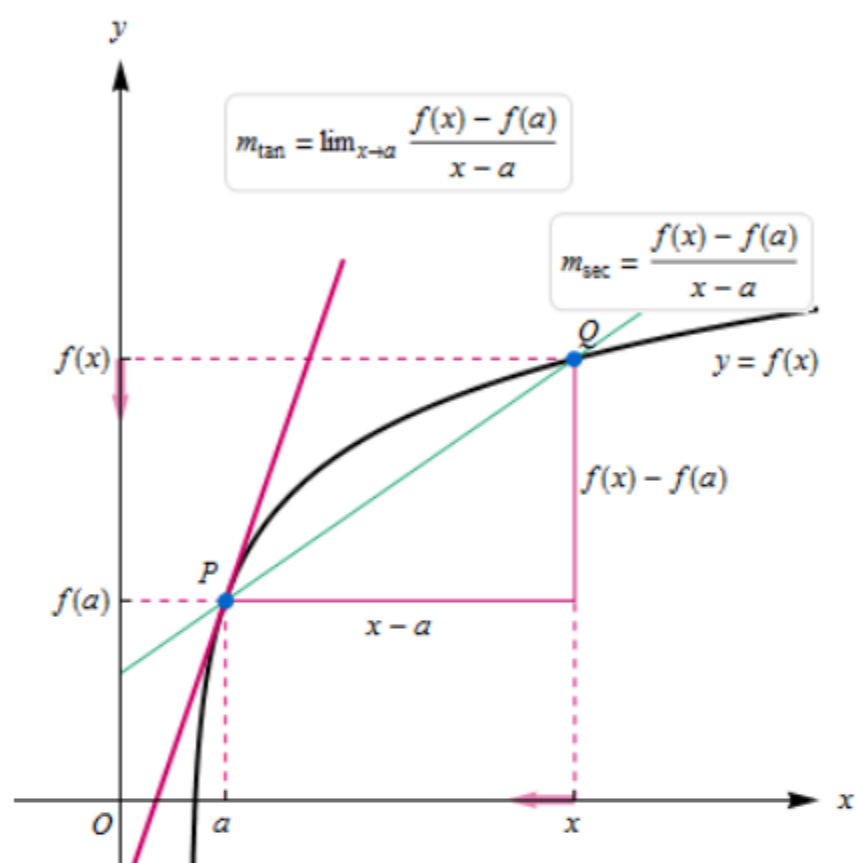
Recitation Questions

Problem 1

February 21, 2026

Problem 1

Problem 1 Consider the following two figures depicting the same graph $y =$



Problem 1

- (a) *In Figure 1 on the left, what are the coordinates of P and Q ? In Figure 2 on the right, what are the expressions for the coordinates of P and Q ?*
- (b) *Express the slope of the secant line through P and Q in terms of the above coordinates for each Figure 1 and Figure 2.*
- (c) *Express the slope of the tangent line at the point P in terms of the above coordinates for each Figure 1 and Figure 2.*
- (d) *What is the difference?*
- (e) *For each of the two graphs, which lines are the secant lines?*
- (f) *For each of the two graphs, which lines are the tangent lines?*

Problem 2

February 21, 2026

Problem 2

Problem 2 For each of the following functions find an equation of the tangent line at the given point.

(a) $f(x) = -5x^2 + 7x - 9$ at $x = 3$.

(b) $g(u) = \sqrt{5u - 4}$ at $u = 3$.

(c) $s(z) = \frac{z}{z - 5}$ at $z = 3$.

Problem 3

February 21, 2026

Problem 3

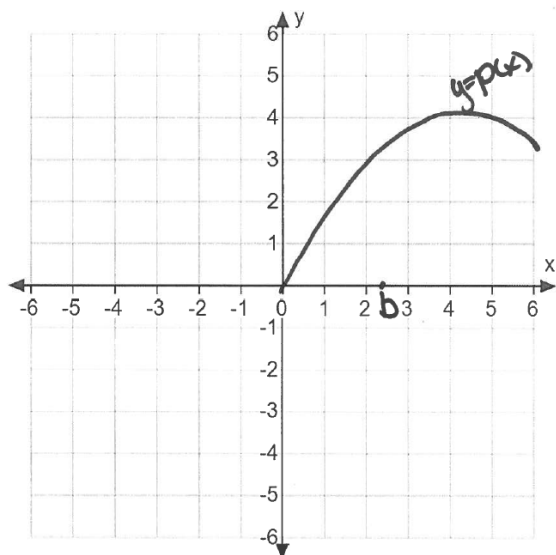
Problem 3 Find an equation of the tangent line at the given point. Then graph the function and the tangent line on the same plot. $f(x) = \sqrt{x+1}$ at $x = 3$

Problem 4

February 21, 2026

Problem 4

Problem 4 The graph of a function p and a point in its domain, b , are shown in the figure below.



- (a) In the figure above, draw and mark clearly the quantity $\Delta y = p(b) - p(1)$ and the quantity $\Delta x = b - 1$.
- (b) Complete the sentence. The quotient $\frac{p(b) - p(1)}{b - 1}$ is the slope...
- (c) Complete the sentence. Provided it exists, the limit $\lim_{x \rightarrow 1} \frac{p(x) - p(1)}{x - 1}$ is the slope...

Problem 5

February 21, 2026

Problem 5

Problem 5 *An object moving along a straight line has a position given by $s(t) = \frac{1}{t-4}$, where s is measured in meters and t in seconds. Find the velocity of the object at time $t = 6$.*